AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 16 · STUDENT LESSON

Accuracy and Imbalance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Keep the class counts. Show the missed cases. State the priority.

Compare ordinary accuracy, equal-class summaries and error consequences under explicit evidence.

Keep one habit: Complete matrices → class fractions → pictured18/2contrast → supplied mix/cost simulation → independent100-case comparison.

MAIN LESSON · SECTION 1

Which class does the headline accuracy hide?

> Report the class counts and error pattern before choosing a metric.

Learning goals

I can explain why overall **accuracy** can hide failure on a less common class.

I can compare a **training-majority baseline** with class-specific results.

I can calculate an equal-class average and report its limits alongside the confusion matrix.

Retrieve labelled confusion-matrix reasoning

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.
- With reference rows and **prediction** columns ordered clear then blocked, where are false negatives when blocked is positive?

End goal

By the end, I can explain **accuracy** and imbalance using a traced example and a stated limitation.

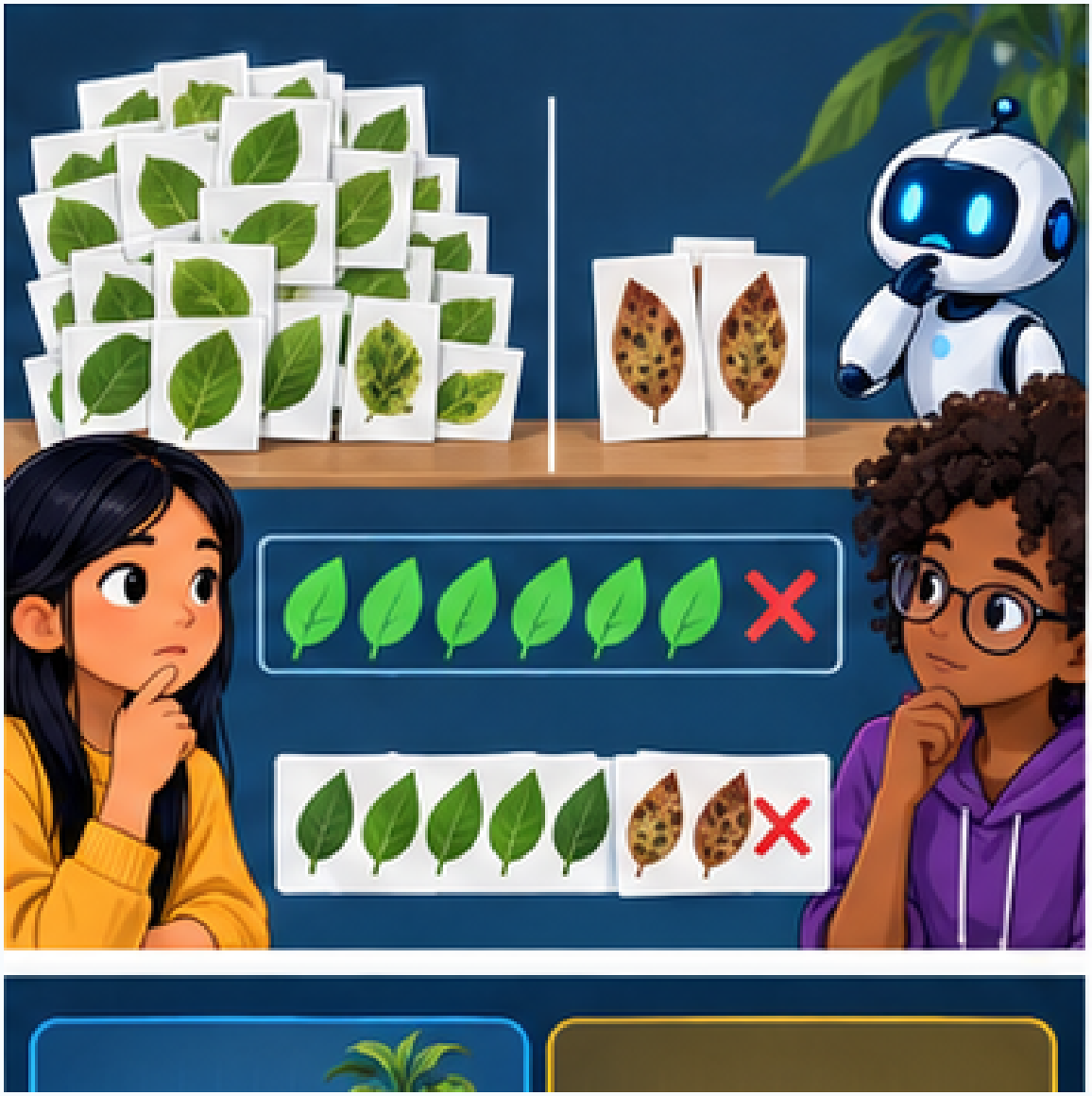
▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Accuracy can be dominated by a common class and hide failure on a rare but important class.



LEARNING TARGET Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Compare two methods on the same imbalanced evaluation set

> The baseline constant comes from training, not evaluation labels.

Accuracy and fictional source contract

Accuracy counts all correct **prediction**s and divides by all evaluated examples. Each example has equal weight, so a common class contributes more to the score. A high result can hide failure on a rare class.

This week keeps the confusion matrix visible and checks both classes separately.

New fictional **training** data has 16 clear and 4 blocked labels. Its **majority baseline** predicts clear for every query; choose that rule from **training**, not evaluation labels.

A separate evaluation set also contains 16 clear and 4 blocked cases. A frozen candidate and the baseline are evaluated on the same 20 records.

Complete baseline/candidate matrices

Method and reference row	Predicted clear	Predicted blocked
Baseline reference clear	16	0
Baseline reference blocked	4	0
Candidate reference clear	12	4
Candidate reference blocked	0	4

Read both80%totals and different errors

Rows are references and columns are **prediction**s, ordered clear then blocked. Blocked is positive. Both methods get 16/20=80 percent **accuracy**. The baseline misses all four blocked cases.

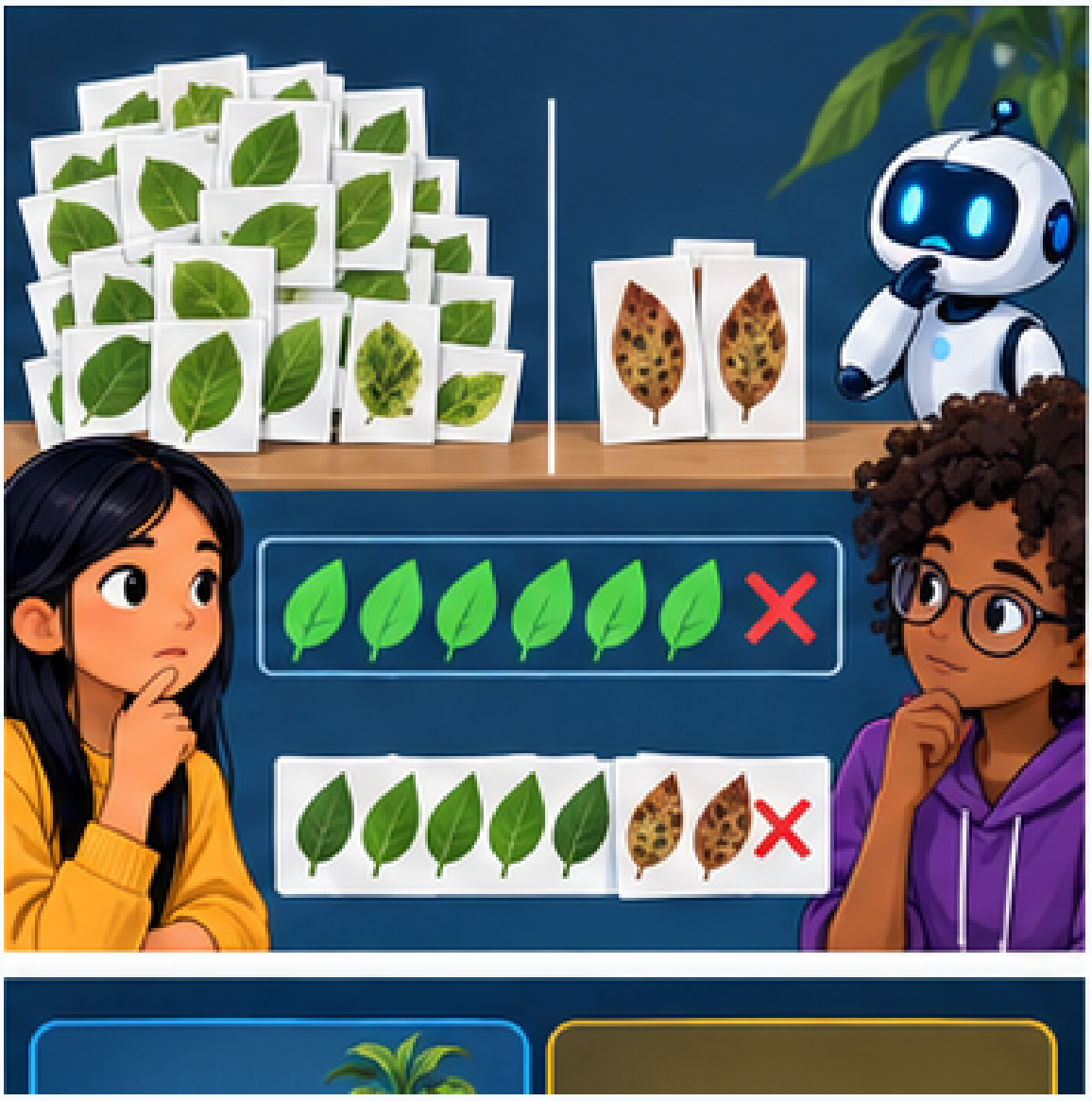
The candidate finds all four but falsely flags four clear cases. The same score conceals a different tradeoff.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **accuracy** and imbalance to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Give each class a visible denominator

> Ordinary accuracy weights cases; balanced accuracy averages the two class recalls equally.

Complete ordinary/balanced score script

```
def scores(counts):
    clear_n = sum(counts[0])
    blocked_n = sum(counts[1])
    if clear_n == 0 or blocked_n == 0:
        raise ValueError("Need evaluation examples of both classes")
    correct = counts[0][0] + counts[1][1]
    accuracy = correct / (clear_n + blocked_n)
    clear_fraction = counts[0][0] / clear_n
    blocked_fraction = counts[1][1] / blocked_n
    balanced = (clear_fraction + blocked_fraction) / 2
    return accuracy, balanced

baseline = [[16, 0], [4, 0]]
candidate = [[12, 4], [0, 4]]
for name, counts in [("baseline", baseline), ("candidate", candidate)]:
    accuracy, balanced = scores(counts)
    print(name, round(100 * accuracy, 1), round(100 * balanced, 1))
```

Trace candidate fractions and equal average

The **output** is baseline 80.0 50.0, then candidate 80.0 87.5. In each **output** line the first percentage is ordinary **accuracy** and the second is **balanced accuracy**.

For the candidate, the clear-class correct fraction is 12/16=75 percent and the blocked-class fraction is 4/4=100 percent. Their equal average is 87.5 percent.

Trace baseline fractions and absent-class policy

For the baseline, those fractions are 16/16=100 percent and 0/4=0 percent, averaging 50 percent. The code assumes a two-by-two table of nonnegative integer counts in the stated order.

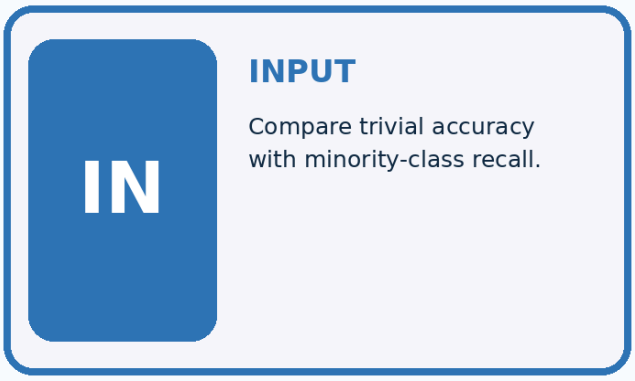
It stops when either reference class is absent. Then report the observed counts and the unavailable class result; do not invent a denominator or silently substitute zero.

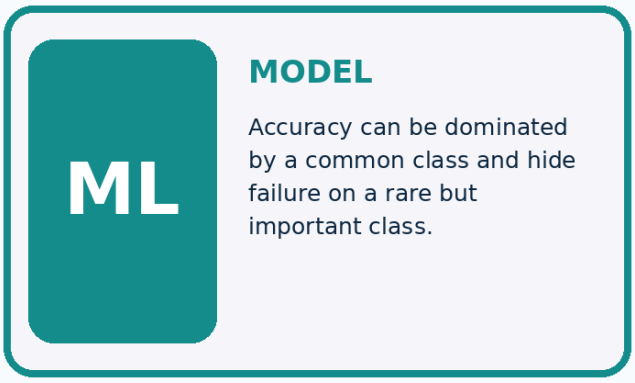
▼ Earlier wording, alternatives and picture transcripts

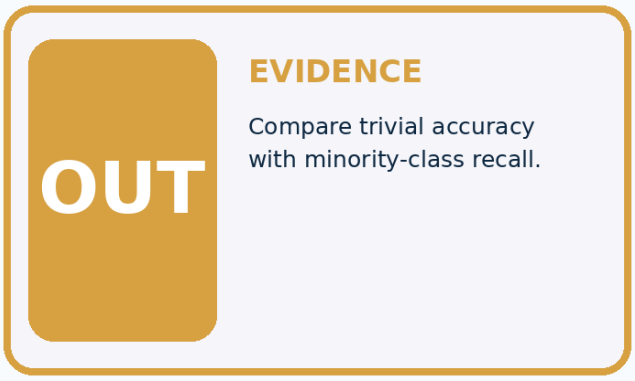
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Compare trivial **accuracy** with minority-class **recall**.

MODEL IDEA **Accuracy** can be dominated by a common class and hide failure on a rare but important class.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Test set has 95 healthy and 5 diseased leaves. Always predicting healthy gives 95 ÷ 100 = 95% **accuracy**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

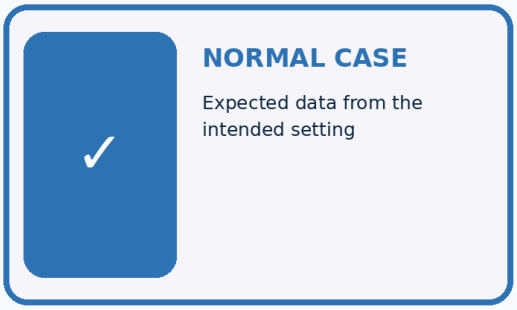
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

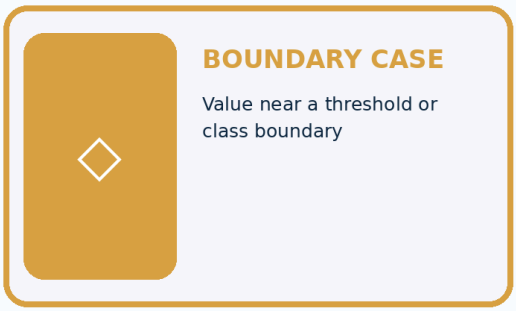
Compare a large majority score with rare-class detection

> A metric can describe a tradeoff without deciding its consequences.

Four evaluation conditions



NORMAL Expected data and expected behavior.

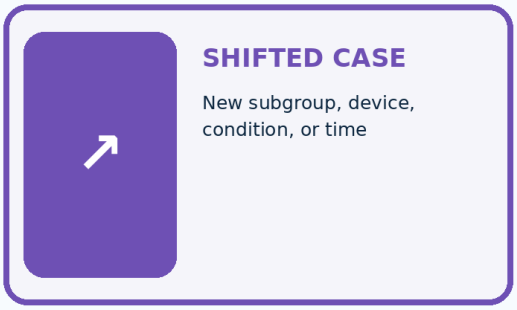


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The candidate’s three false alarms are corrected without changing its detections.

▼ **Check after predicting**

Prediction: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

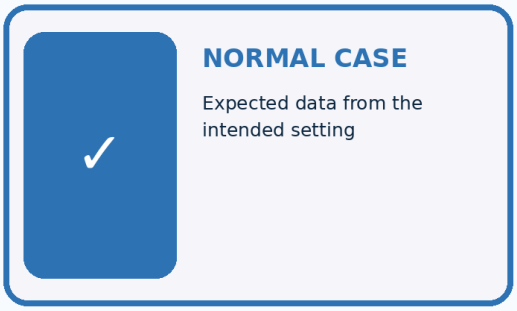
Reason: False alarms move from FP to TN; neither TP nor FN changes.

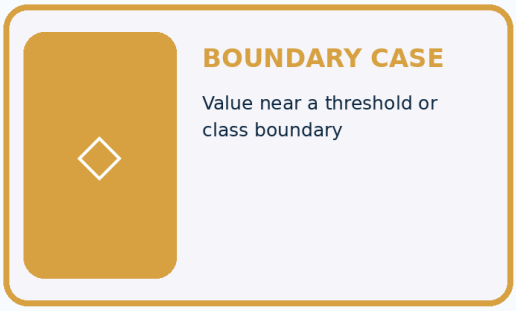
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

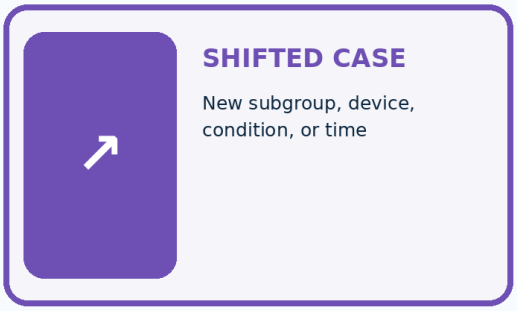
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the rare-class failures beside the full matrix

> Use the picture’s own counts rather than the main corridor denominators.

Complete pictured facts and scope

Show how a common healthy class can hide complete failure to detect spotted leaves.

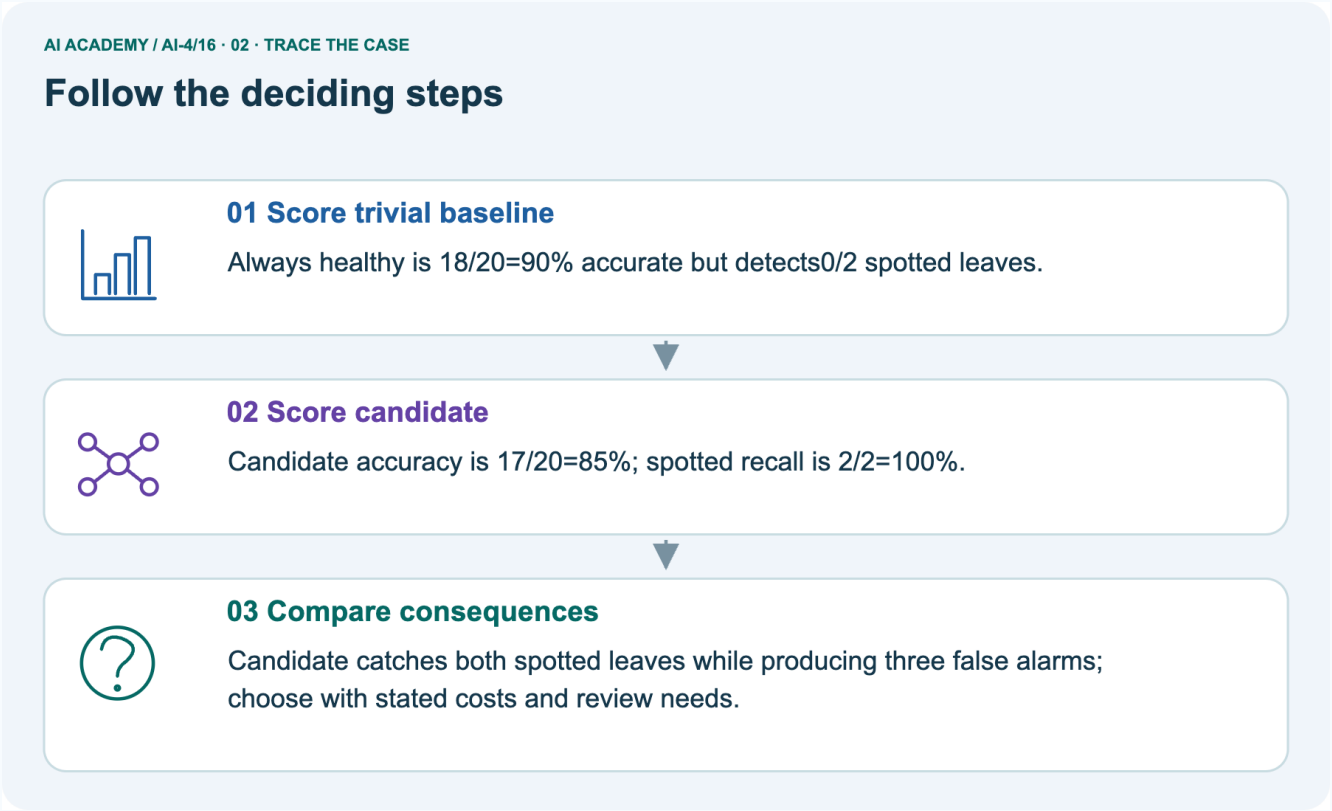
Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy. A candidate hasTP2, FN 0, FP 3, TN 15.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main corridor fixture has16clear/4blocked: baseline and candidate both16/20accuracy, but blocked recall0/4versus4/4. Main candidate false alarms4. Pictured leaf fixture instead has18healthy/2spotted: always-healthy18/20accuracy,0/2spotted detected; candidate TP2,FN0,FP3,TN15 gives17/20accuracy and2/2spotted recall.

A hypothetical review correcting all3false alarms gives20/20reviewed outcomes; preserve the original model-only17/20record. Neither two spotted examples nor a reviewed perfect score establish reliable unseen-case detection.

Trace accuracy, missed cases and false alarms



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

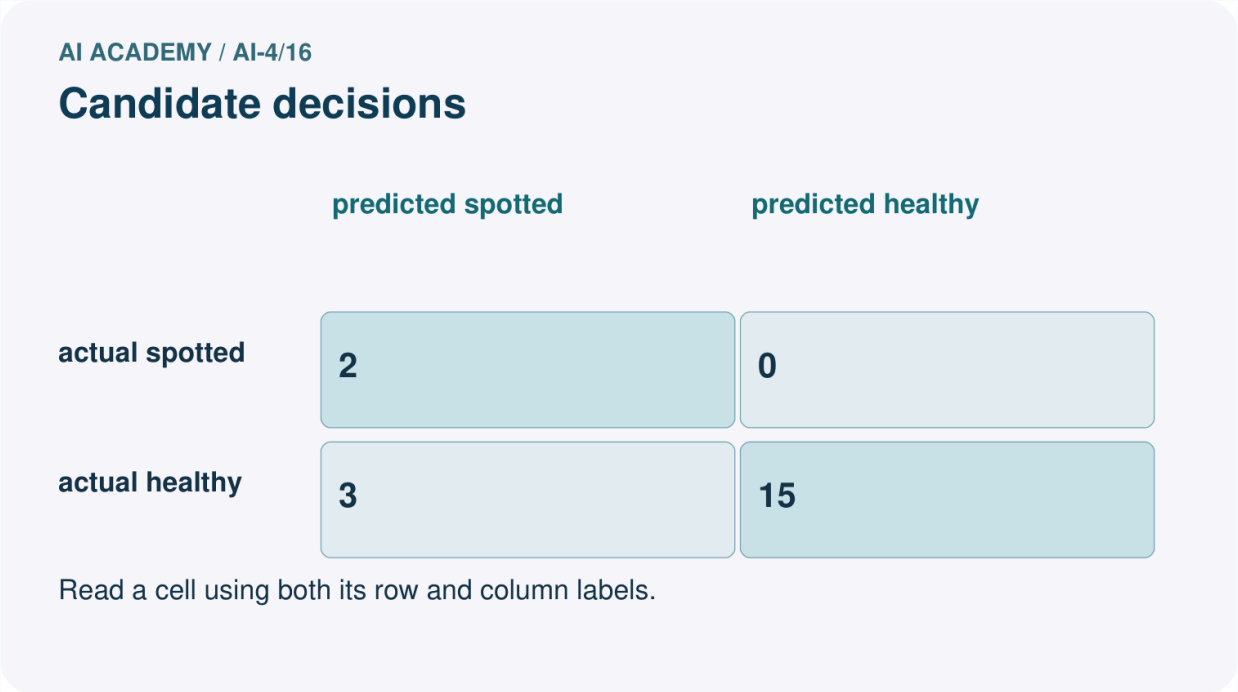
STEP	WHAT HAPPENS AND WHY
1. Score trivial baseline	Always healthy is 18/20=90% accurate but detects0/2 spotted leaves.
2. Score candidate	Candidate accuracy is 17/20=85%; spotted recall is 2/2=100%.
3. Compare consequences	Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the pictured candidate’s labelled matrix

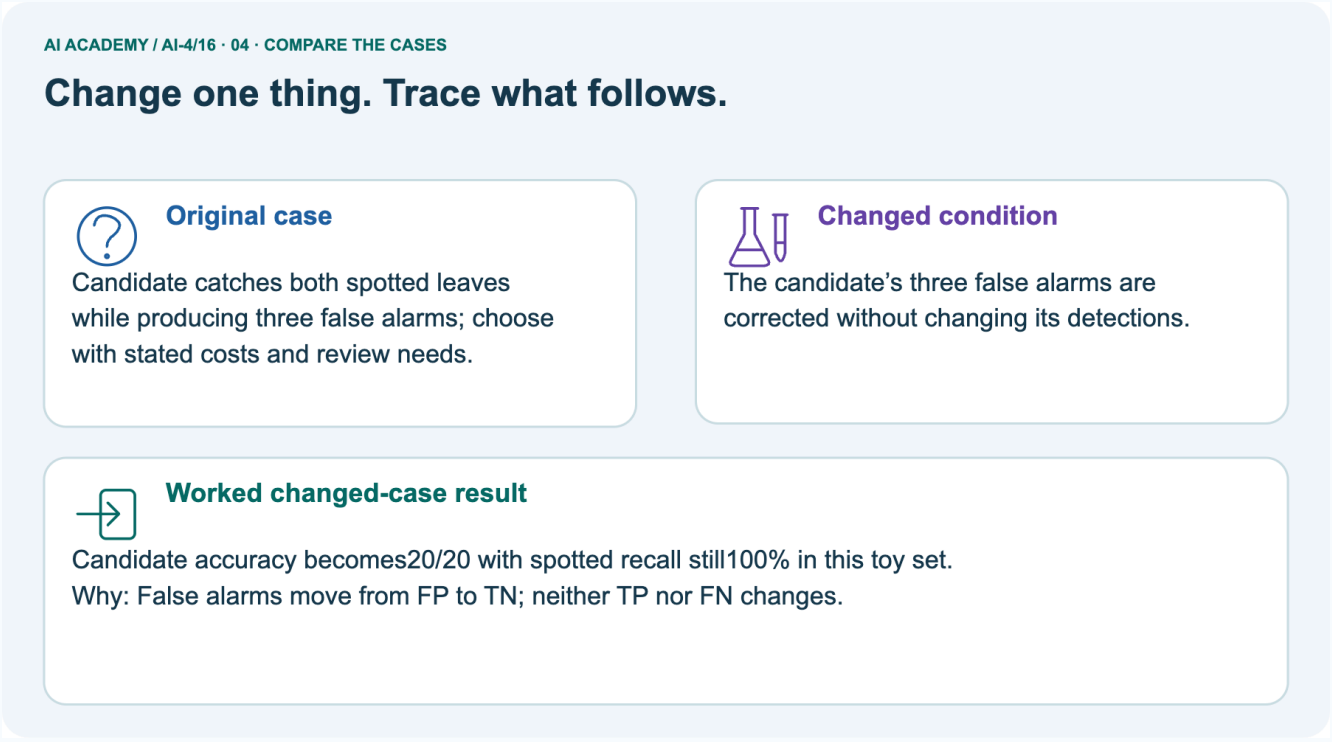


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict correcting only the three false alarms

- Change one thing: The candidate’s three false alarms are corrected without changing its detections. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State evidence and task limits

- Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

- Read the labelled picture: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main equal-accuracy comparison

Same overall score different class results

SUMMARY	ALWAYS CLEAR BASELINE	FROZEN CANDIDATE
Overall correct	16 of 20 is 80 percent	16 of 20 is 80 percent
Reference clear correct	16 of 16 is 100 percent	12 of 16 is 75 percent
Reference blocked correct	0 of 4 is 0 percent	4 of 4 is 100 percent

Overall accuracy weights classes by their number of evaluation examples. Balanced accuracy gives the two class-specific fractions equal weight.

Both methods score 80 percent. Explain which cases the baseline fails, how the candidate changes the errors, and why their balanced accuracies differ.

Balanced accuracy is a metric choice, not a repair to the classifier or a guarantee of safety. It gives classes equal influence in this summary; it does not encode the actual consequences of false alarms and missed cases.

Four blocked evaluation examples are also too few for a strong general claim.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 16 • WORKED EVIDENCE

1. Test set has 95 healthy and 5 diseased leaves.
2. Always predicting healthy gives 95 ÷ 100 = 95% accuracy.
3. It finds 0 of 5 diseased leaves, so diseased recall = 0%.
4. Report class counts and recall; redesign data or use only where misses are acceptable.

STEP 1 Test set has 95 healthy and 5 diseased leaves.

STEP 2 Always predicting healthy gives 95 ÷ 100 = 95% **accuracy**.

STEP 3 It finds 0 of 5 diseased leaves, so diseased **recall** = 0%.

STEP 4 Report class counts and **recall**; redesign data or use only where misses are acceptable.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

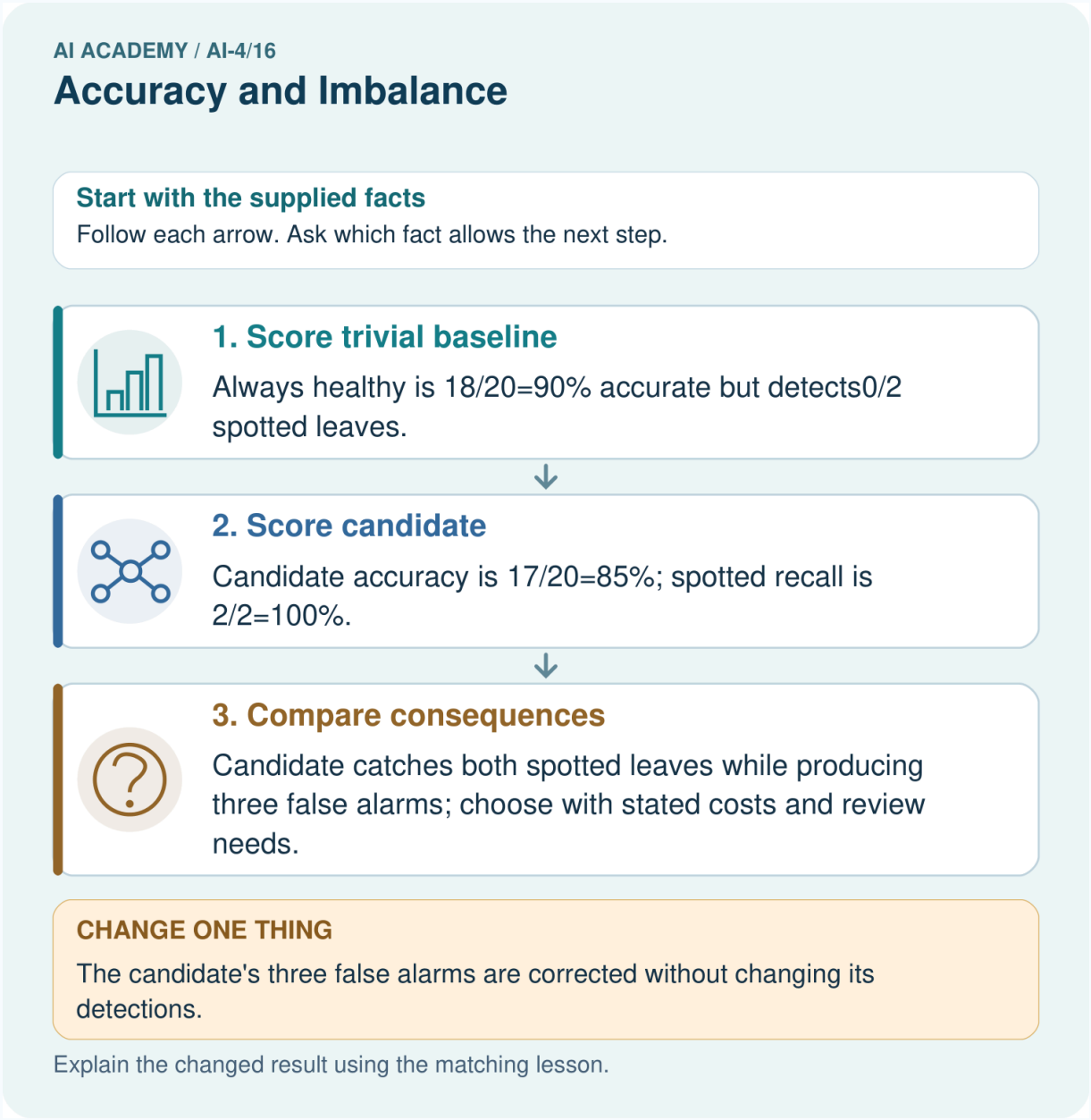
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Accuracy and Imbalance

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find an overall score used as proof for every class

> A summary metric changes what it emphasizes; it does not repair predictions.

Diagnose the main claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A classifier with 95% **accuracy** is excellent in every practical situation.

Explain the interpretation and limits

- Compare methods on the same appropriate evaluation examples and report the **training**-derived baseline. Include ordinary **accuracy**, each class denominator, false positives and false negatives.

  An equal-class summary adds information but cannot choose the acceptable tradeoff for the task. Keep development decisions separate from a final test.

- A result of 4/4 correct blocked cases is 100 percent on those four examples, not proof of perfect future detection. If the evaluated set contains no blocked cases, it cannot measure blocked-case performance at all.

  Next week separates precision and **recall** by asking which denominator answers each question.

> **A team advertises 80 percent accuracy but its classifier predicts clear on every record. What additional evidence would you request, and what can you already conclude about the four blocked cases in our worked evaluation?**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A classifier with 95% accuracy is excellent in every practical situation. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Practise both class fractions and a changed class mix

> Do not treat an arithmetic population example as a measured distribution-shift result.

Try W2 from the complete matrices

Use workbook W2 for the connected example "**Accuracy** and **class imbalance**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Trace the separate8clear/8blocked illustration

Consider a separate illustrative set with 8 clear and 8 blocked cases. Suppose the same within-class fractions hold: the baseline matrix is [[8,0],[8,0]] and the candidate matrix is [[6,2],[0,8]].

Ordinary **accuracy** is now 50 percent versus 87.5 percent. **Balanced accuracy** remains 50 percent versus 87.5 percent. The class mix changed the overall score even though the within-class fractions stayed fixed.

This is an arithmetic comparison, not proof that a real classifier keeps the same behavior after a population change. Report sample size, class counts, full matrix and how the evaluation data was chosen.

Do not remove inconvenient cases or duplicate rare evaluation cases just to improve a headline number. If **training** is resampled, keep that separate from evaluation and document it.

Challenge high-score-means-both-classes reasoning

- Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured detection failure

- Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the new100-case task

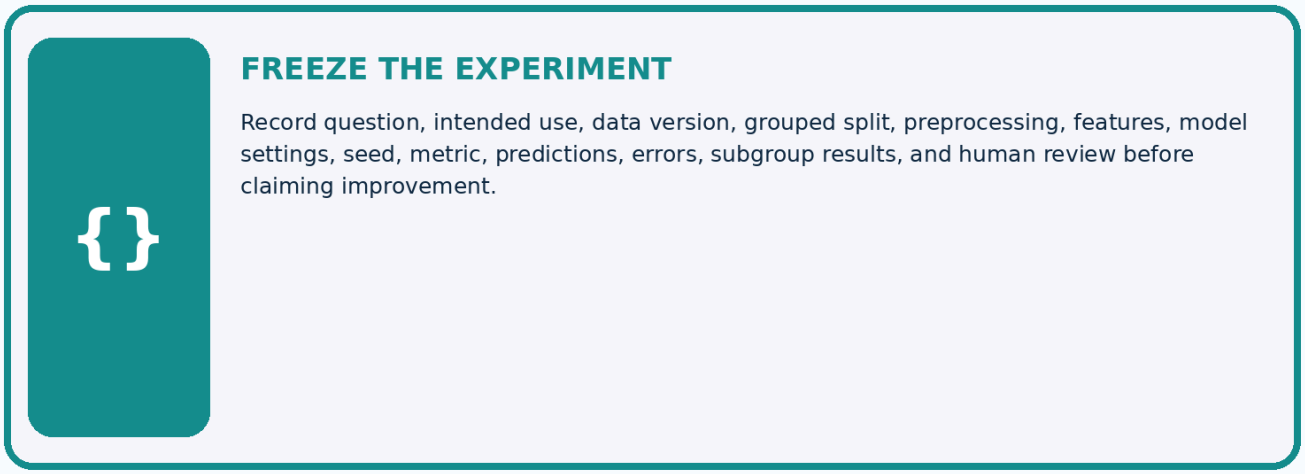
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Compare both class fractions

Check your reasoning

Explain the picture

Run a complete imbalance and consequence comparison

> The task supplies every matrix and both cost policies.

Complete main and changed-mix matrices

Complete matrix fixture, clear/blocked axes	Baseline	Candidate
Main16clear/4blocked	[[16,0],[4,0]]	[[12,4],[0,4]]
Changed class mix8clear/8blocked,within-class fractions held equal by construction	[[8,0],[8,0]]	[[6,2],[0,8]]
Ordinary accuracy, changed mix	8/16=50%	14/16=87.5%
Balanced accuracy, both constructed mixes	(1+0)/2=50%	(0.75+1)/2=87.5 %

Two explicit fictional cost policies

Declared fictional classroom cost policy	Main baseline FN4/FP0	Main candidate FN0/FP4
Policy A: missed blocked costs5points;false alarm1point	4×5=20points	4×1=4points
Policy B: missed blocked costs1point;false alarm6points	4×1=4points	4×6=24points
Meaning	Different declared priorities can change which toy error pattern is preferred.	These arbitrary points are supplied for arithmetic, not measured real-world harms or an authorized deployment decision.

Normal, sparse, absent-class and changed-mix probes

Complete score probe	Expected bounded check
Normal main candidate [[12,4],[0,4]]	Ordinary80%;balanced87.5%;FP4,FN0.
Boundary rare class has one case: [[3,1],[1,0]]	Ordinary3/5=60%;class recalls3/4and0/1;balanced37.5%,very sparse positive evidence.
Missing positive class: [[4,0],[0,0]]	Observed ordinary accuracy4/4, but no blocked recall or two-class balanced result under this lesson's policy; original scores function stops.
Changed class mix supplied above	Ordinary accuracy changes with class proportions; held-fixed class fractions keep balanced accuracy unchanged by construction, not as a promised real model behavior.

Calculate, compare and state the policy

1. Imbalance Simulation: calculate ordinary accuracy, both class recalls and their equal average for the complete main matrices. List every kind of error before applying the two predeclared cost policies. Then repeat the arithmetic for the supplied8/8class-mix fixture, keeping the constructed within-class behavior explicit.

   Use the original complete Python scores program for nonempty-class matrices; record its absent-class stop honestly. Do not duplicate evaluation cases or change denominators to improve a score. Keep the independent90healthy/10spotted comparison for an uncoached first attempt.

Retain evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____

> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.
</details>

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Imbalance Simulation: compare majority prediction with a model using accuracy, recall, and the actual error costs.

1. Write the hypothesis and expected **evidence** before running.

2. Imbalance Simulation: compare majority **prediction** with a model using **accuracy**, **recall**, and the actual error costs.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare accuracy with spotted detection on fresh counts

> Show both denominators and state the false-alarm tradeoff.

Complete selected counts and scope

Validation has90 healthy and10 spotted cases. Always-healthy predicts all healthy. Another model has TP7,FN3,FP8,TN82. Compare **accuracy** and spotted **recall**.

This selected task has90healthy/10spotted and its own candidate TP7,FN3,FP8,TN82. Use spotted as the declared positive class and retain these100cases as the denominator. No actual consequence weights or broad reliability evidence are supplied; give conditional reasoning rather than declaring a universally best model. Do not substitute the20-case main or pictured matrices.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classifier predicts healthy for all 100 plants; 95 are healthy and 5 diseased. Compute **accuracy** and diseased **recall**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the rare class visible in the project record

> A higher chosen metric alone cannot justify a use decision.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

Keep class counts, training-derived baseline, full evaluation matrix, ordinary accuracy, class-specific recalls and any balanced summary with denominators. State the declared consequences and review policy separately from the metric. Missing classes and tiny positive samples limit the evidence; if training is resampled, document it without altering the evaluation population or hiding mistakes.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits, uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

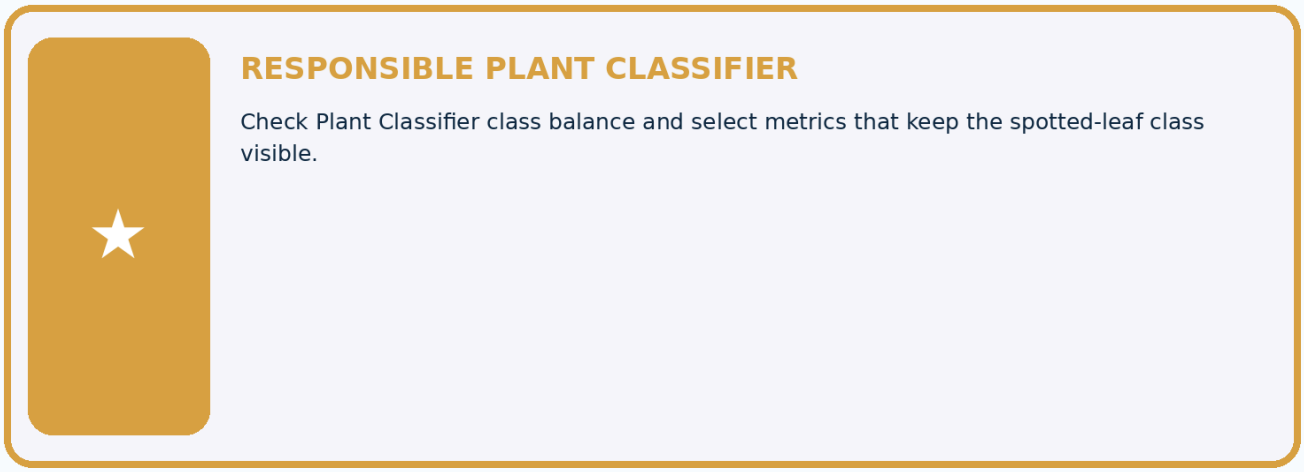
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with a headline score and the failures behind it

> Explain which denominator and consequence each conclusion uses.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain why overall **accuracy** can hide failure on a less common class

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
accuracy	correct predictions divided by all predictions	
class imbalance	classes occur in very different amounts	
minority class	less frequent target class	
recall	fraction of actual positives found	

Closer-explanation terms

Word	Meaning
Class imbalance	Unequal numbers of examples in different classes
Majority baseline	A rule that always predicts the most common training label

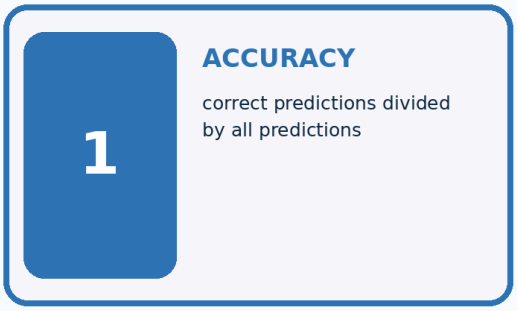
Word	Meaning
Class-specific correct fraction	Correct predictions for one reference class divided by the number of examples of that class
Balanced accuracy	The average of the class-specific correct fractions with equal weight for each class

▼ Earlier wording, alternatives and picture transcripts

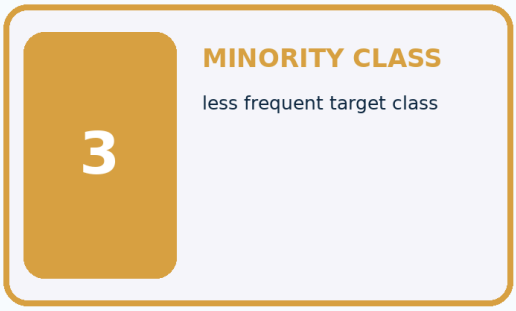
VOCABULARY

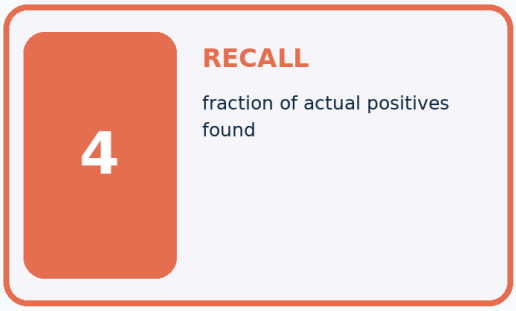
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A classifier with 95% **accuracy** is excellent in every practical situation.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **accuracy**, **class imbalance**, **minority class**, **recall**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A classifier with 95% **accuracy** is excellent in every practical situation.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

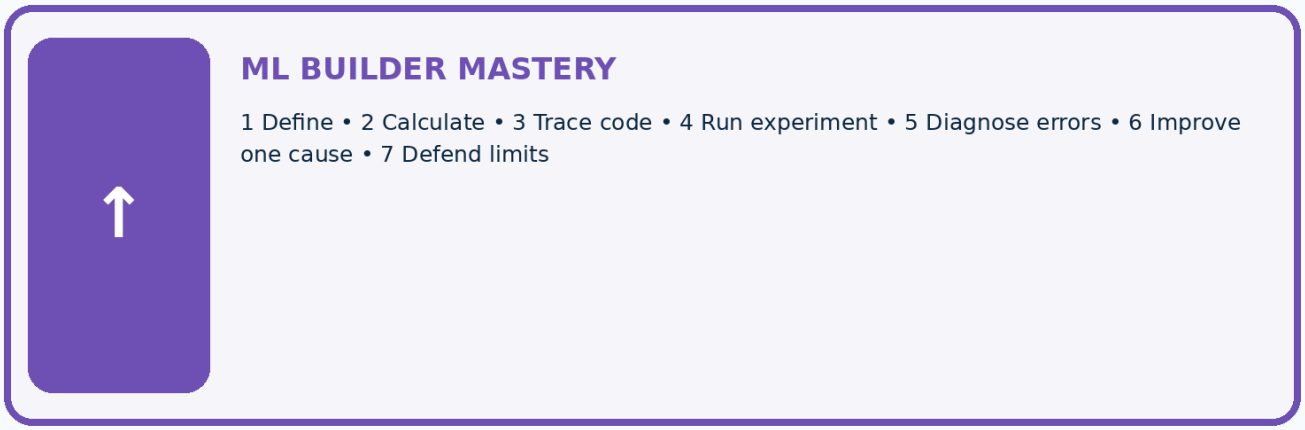
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh90healthy/10spotted comparison for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

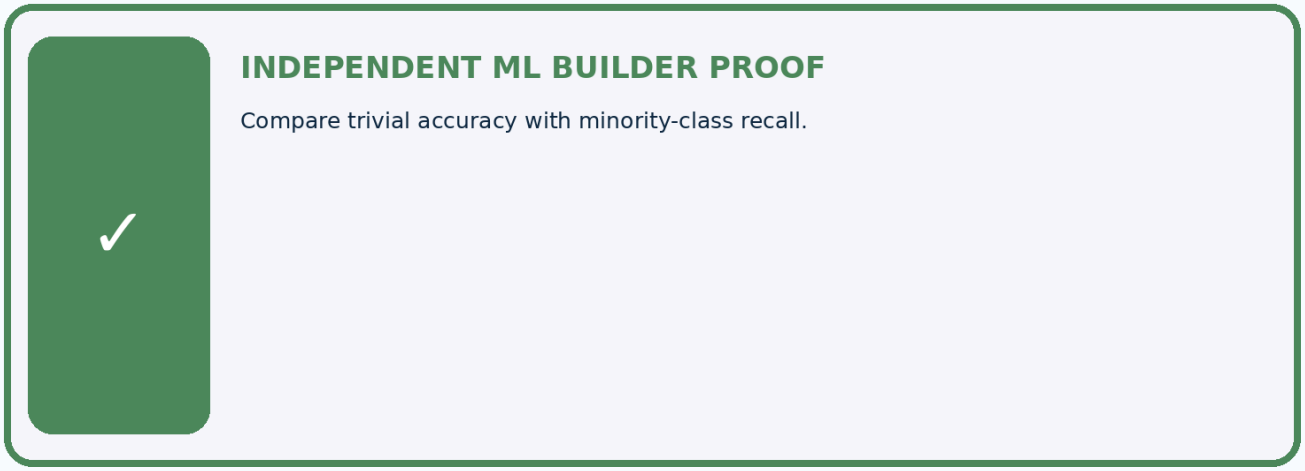
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Compare trivial **accuracy** with minority-class **recall**.

NEXT BRIDGE Carry the experiment **evidence** into: Precision and **Recall**.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.

Calculate and critique **accuracy**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

95/100=95%, but spotted **recall**=0/5=0%. Majority performance hides the detection failure.

Evidence or stage	Value or result
Healthy, predicted healthy	95
Spotted, predicted healthy	5

▼ Earlier wording, alternatives and picture transcripts

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

The majority can hide a detection failure

A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.

Calculate and critique **accuracy**.

95/100=95%, but spotted **recall**=0/5=0%. Majority performance hides the detection failure.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Explain why overall accuracy can hide failure on a less common class	Explain the deciding step, show a trace or test, and state one limit.
Compare a training-majority baseline with class-specific results	Explain the deciding step, show a trace or test, and state one limit.
Calculate an equal-class average and report its limits alongside the confusion matrix	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Show how a common healthy class can hide complete failure to detect spotted leaves.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Show how a common healthy class can hide complete failure to detect spotted leaves.

- Which labelled step matches this detail? Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.
- 1. Compare consequences
- 2. Score candidate
- 3. Score trivial baseline
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare consequences Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. Read the picture in order.

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The candidate's three false alarms are corrected without changing its detections. Which conclusion is supported by the supplied facts?

- 1. Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. False alarms move from FP to TN; neither TP nor FN changes.

Repair and keep the record

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training accuracy** replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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<summary>Reference · The six-picture journey</summary>

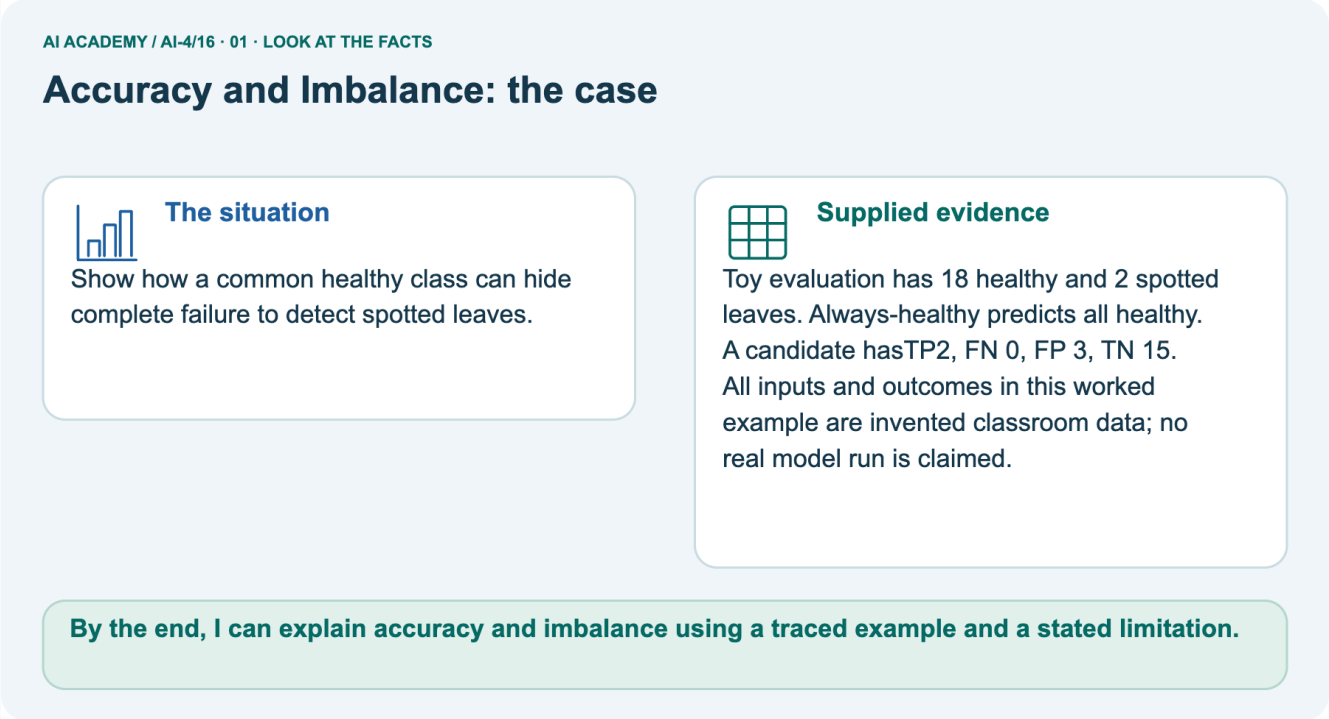
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Accuracy and Imbalance: the case



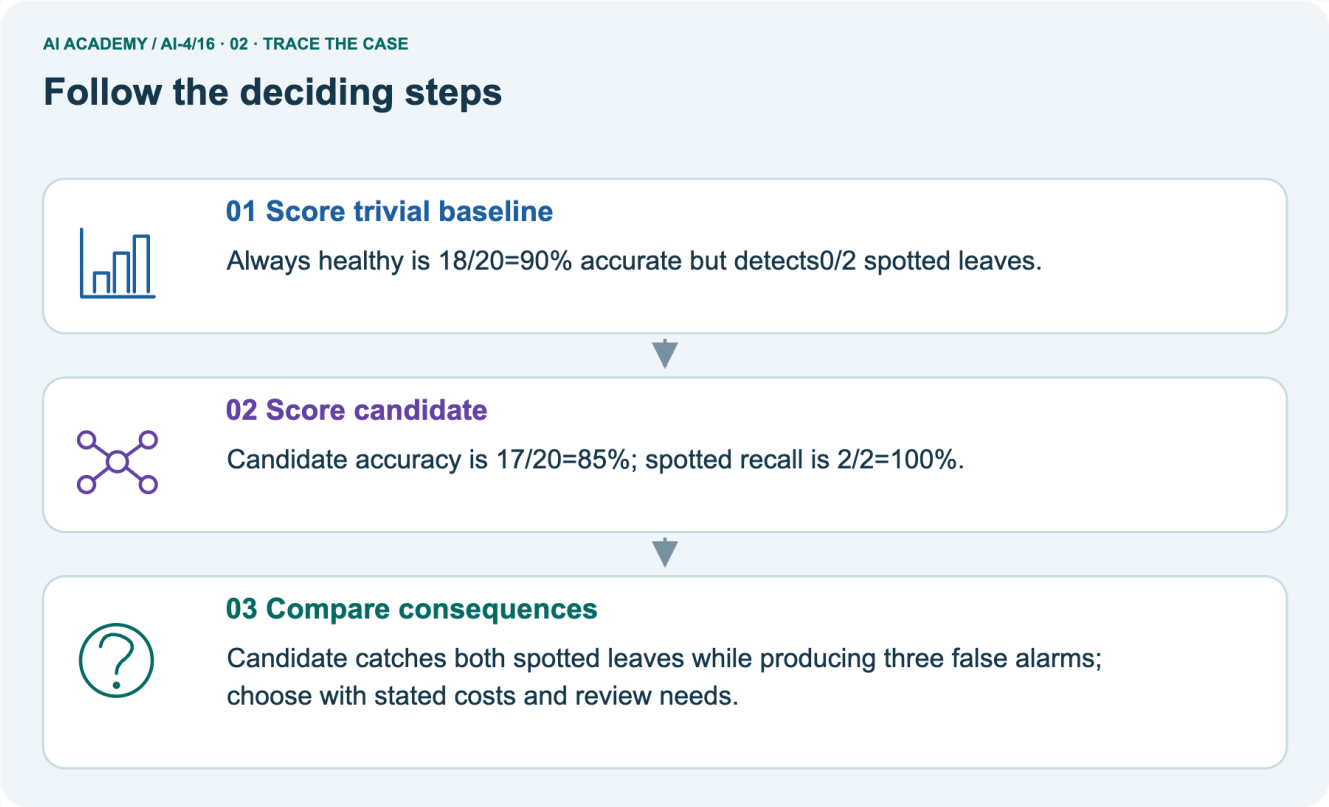
Follow the labelled picture with the matching explanation. Open the vector diagram

Show how a common healthy class can hide complete failure to detect spotted leaves.
Supplied facts: Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy.

A candidate hasTP2, FN 0, FP 3, TN 15. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

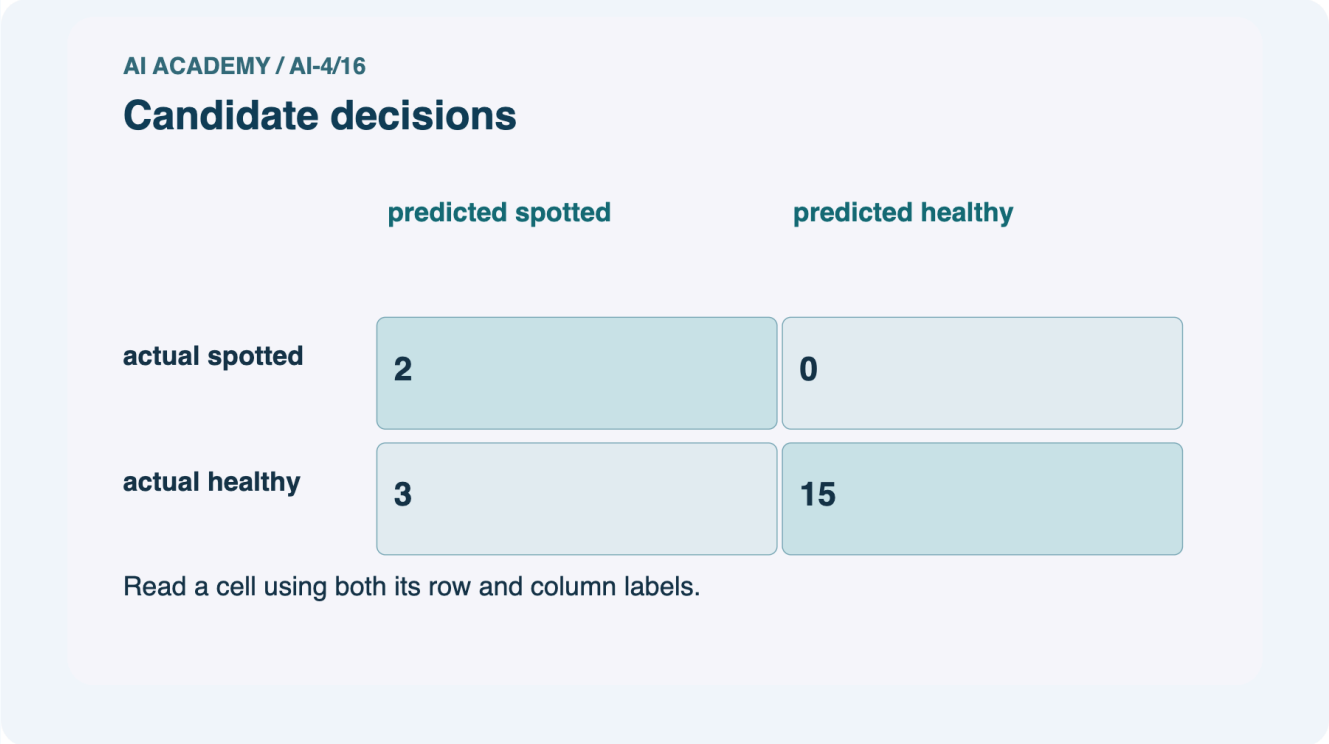


Follow the labelled picture with the matching explanation. Open the vector diagram

Score trivial baseline: Always healthy is 18/20=90% accurate but detects0/2 spotted leaves. → Score candidate: Candidate **accuracy** is 17/20=85%; spotted **recall** is 2/2=100%. → Compare consequences: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Point to each numbered stage and explain which recorded **input** it uses.

03. Candidate decisions

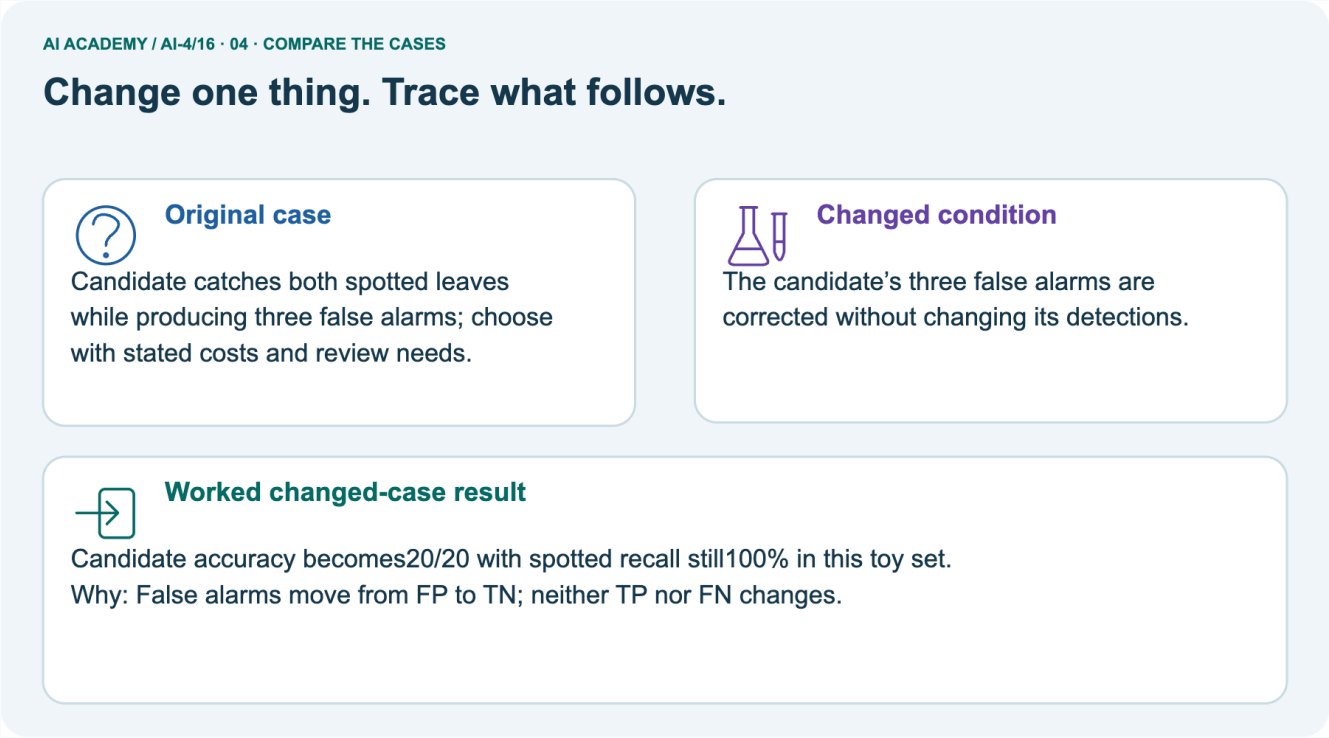


Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate decisions. actual spotted: predicted spotted = 2, predicted healthy = 0. actual healthy: predicted spotted = 3, predicted healthy = 15.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

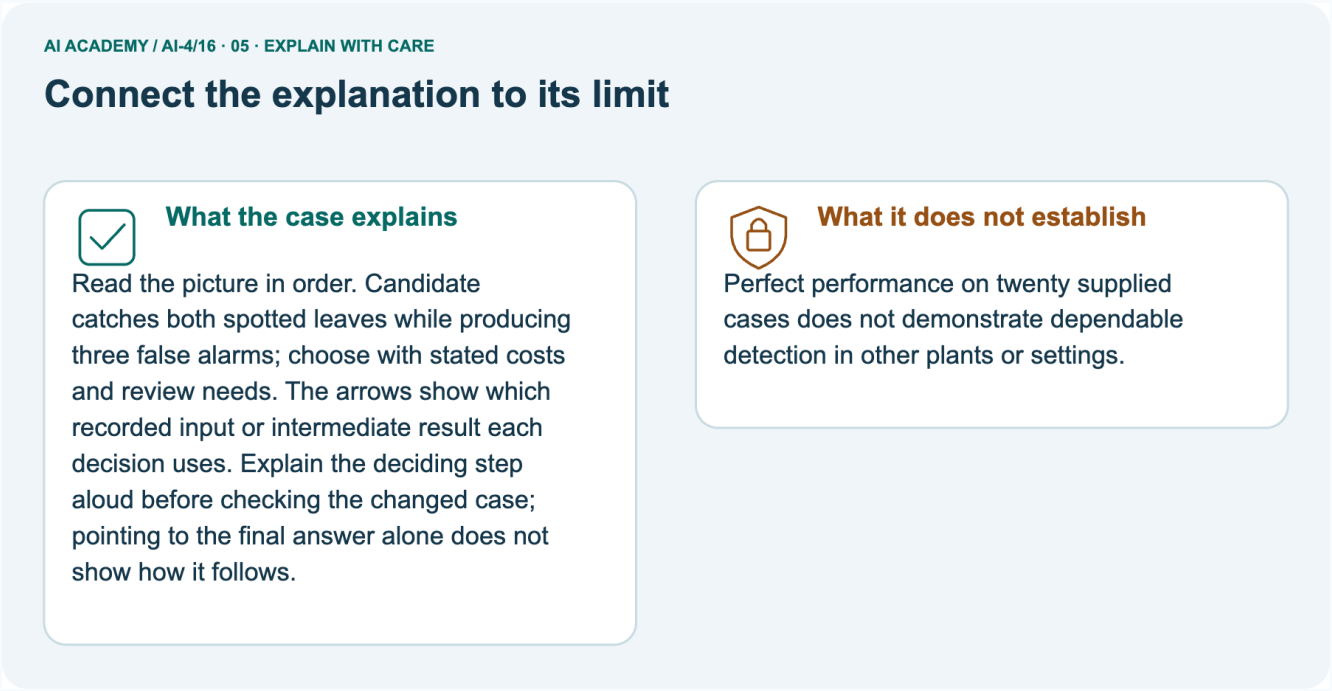
Original case: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Changed condition: The candidate's three false alarms are corrected without changing its detections. Expected result: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

Reason: False alarms move from FP to TN; neither TP nor FN changes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

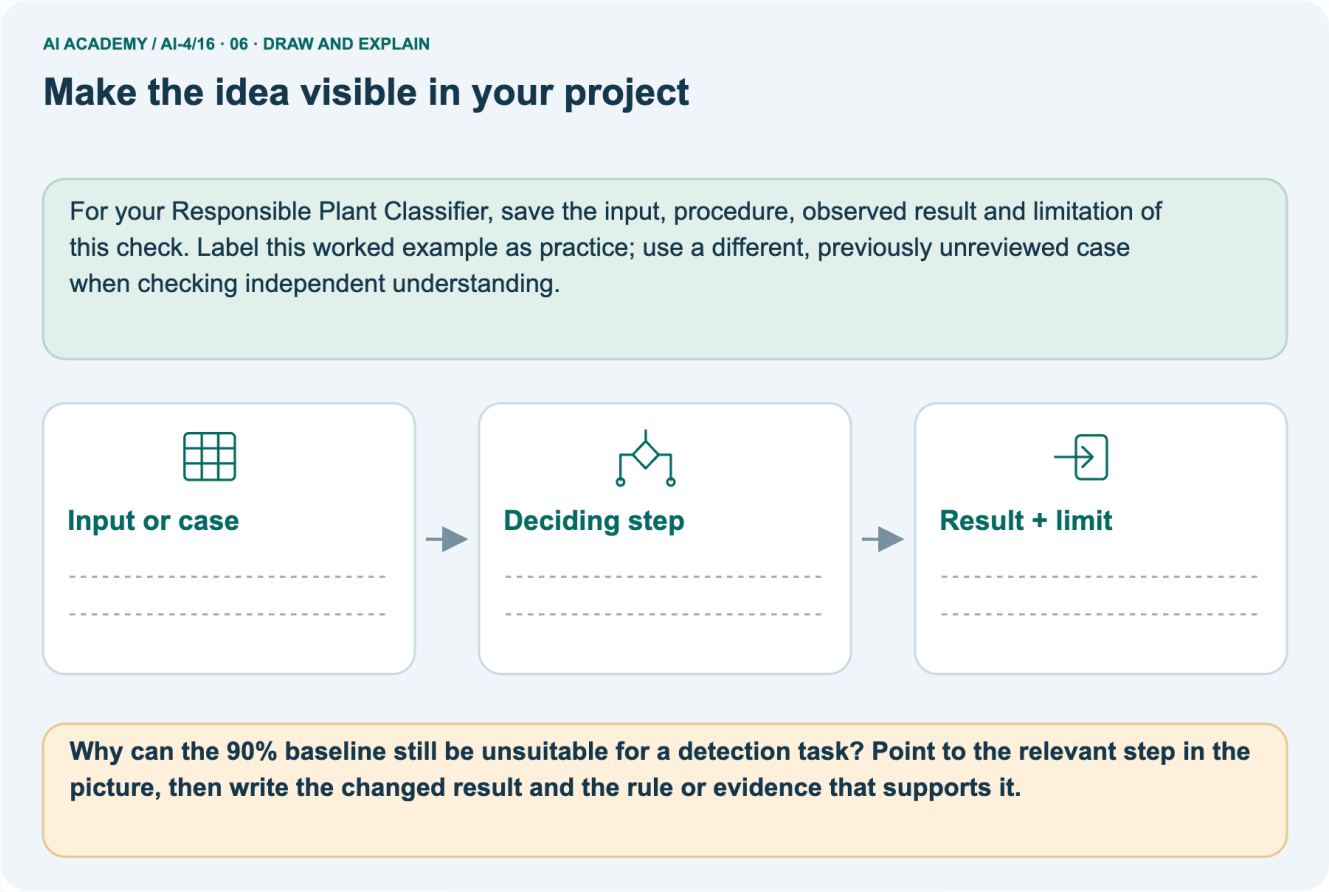
Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Accuracy and Imbalance: the case

02. Follow the deciding steps

03. Candidate decisions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.